

142 Questions, 142 Answers

NexGen iMATCH — har sawal ka ek-line jawab, role ke hisaab se

Section	Kaun poochhega	Q#
A	Forensic investigator / examiner	1–28
B	Legal / procedural reviewer	29–40
C	Database engineer	41–58
D	Software / backend engineer	59–81
E	Security engineer	82–95
F	AI / ML engineer	96–125
G	Product / management	126–135
H	Killer questions — inhe sabse pehle ratto	136–142
—	Architecture layer cheat sheet	Annex

Paanch jawab jo sabse zyada dohrane padenge

1. Output **lead** hai, identification nahi. **2.** Faisla hamesha **insaan** ka — engine me *confirmed* likhne ka code path hi nahi hai. **3.** Model **pretrained ArcFace** hai; hamara contribution measurement discipline hai. **4.** Sab kuch ek **6 GB laptop GPU** par bana — wahi har technical choice ka constraint tha. **5. Koi independent validation nahi hui** — aur ye hum khud pehle batate hain.

Golden rule

Jo limitation wo dhoondh sakta hai, wo aap **pehle** bol do. Investigator over-claiming pakadne ke liye trained hota hai — jab aap khud kami bataoge, baaki har baat par uska bharosa badh jayega.

A. Forensic Investigator / Examiner

Sabse zyada sawal yahin se aayenge. Q1, Q2, Q9 aur Q24 sabse important.

1. Ye system exactly karta kya hai?

Do chevron ko compare karta hai (1:1) aur ek enrolled gallery me search karta hai (1:N) — output ek ranked candidate list hai, lead ke roop me.

2. Iska output evidence hai ya lead? ■

Lead. Evidence sirf human examiner ka conclusion hai, jo report me uske naam ke saath jata hai.

3. Original photo ke saath chhed-chhaad hoti hai kya?

Nahi — original write-once store me jati hai aur har operation uski copy par hota hai.

4. Agar photo badal di jaye to pata chalega?

Haan — har file ka naam uski SHA-256 hash hai, ek byte badla to mismatch turant pakda jayega.

5. Chain of custody kaise maintain hoti hai?

Saat controls se: original untouched, hash-based naming, DB-to-bytes link, full logging, hash chain, named actor, aur mandatory lawful basis.

6. Kaun-kaun si actions log hoti hain?

Enrolment, search, verify, enhancement, adjudication, export, login, API-key use aur account changes.

7. Log ko koi mita ya badal sakta hai?

Nahi — hash chain us point se aage toot jayegi, aur sequence me gap “removed record” report hoga.

8. Log padh kaun sakta hai?

Apne tenant ka koi bhi authenticated user, apne baare wale entries samet — jo log sirf admin dekh sake, use chupke se misuse karna aasan hota hai.

9. Match confirm kaun karta hai — system ya insaan? ■

Insaan. Engine me *confirmed* likhne ka koi code path hai hi nahi.

10. Examiner ka naam report me aata hai?

Haan — examiner ID, timestamp aur uske notes permanently record hote hain.

11. Search karne ka reason record hota hai?

Haan — *lawful basis* mandatory field hai aur verbatim audit chain me jata hai.

12. Chhe mahine baad koi poochhe “ye search kyun ki” — jawab milega?

Haan — jawab record me hai, kisi ki yaadash me nahi.

13. Report me kya-kya hota hai?

Material examined, comparison plates, methodology, limitations, conclusion, adjudication trail aur signature block.

14. Report me limitations likhi hoti hain?

Haan — har page ke footer par caveat, aur ek poora limitations section jo template se aata hai, generate nahi hota.

15. Jo match nahi hua uski report bhi banti hai?

Haan — non-match ko bilkul same structure milta hai; “why not” ko utni hi rigour di jati hai.

16. Enhanced photo report me ja sakti hai? ■

Sirf tab jab “not evidentiary” label ke saath **aur** original ke bilkul saath ho — warna renderer jaan-boojh kar fail karta hai.

17. Enhanced photo se match karte ho?

Original hamesha primary run hai; enhanced ke saath A/B chalta hai aur dono runs apna source kind record karte hain.

18. Raat ki IR footage me colour kahan se aata hai? ■

Wo AI ka banaya hua hota hai — IR capture me colour hoti hi nahi, aur system report par ye mandatory line likhta hai.

19. Video ke frame ka timestamp galat ho sakta hai?

Haan — P/B frame ke pixels dusre frame se copy hote hain, isliye hum I-frames prefer karte hain aur baaki frames ko unke type ke saath label karte hain.

20. “1080p” likha ho to 1080p hi hoti hai?

Nahi — recorder aksar upscale karke store karta hai; hum spectral band-limit se true resolution naapte hain aur wo stored dimensions ke saath report karte hain.

21. Landmark measurements se identity prove karte ho? ■

Nahi — FISWG aur ENFSI dono ne isse reject kiya hai; hum measurements sirf observations ke roop me likhte hain, conclusion ke grounds ke roop me nahi.

22. Deepfake pakad lete ho? ■

Ek heuristic hai, trained classifier nahi — system apne output me khud *certified: false* likhta hai.

23. Liveness detection certified hai? ■

Nahi — ISO/IEC 30107-3 ke against kabhi evaluate nahi hui; ise examiner ka dhyan prioritise karne ke liye use karo, authenticity conclude karne ke liye nahi.

24. Agar banda database me hai hi nahi to system kya karega? ■

System phir bhi list dega — real surveillance par wo stranger reject nahi kar sakta, aur yahi wajah hai ki human review optional nahi hai.

25. System kabhi galat aadmi dikha sakta hai?

Haan — deployed threshold par lagbhag 1,000 comparisons me 1 false match.

26. Kabhi galat result aaya hai? Kya kiya? ■

Haan — do alag logon ka score 0.2405 aaya aur system ne match kaha; humne threshold hi 0.20 se 0.2871 kar diya aur cost measure karke record ki.

27. Kitne candidates return hote hain aur kis order me?

Top-K, cosine similarity ke descending order me, subject-collapsed — ek hi vyakti ki kai templates ek entry ban jati hain.

28. Examiner ko training chahiye is system ke liye?

Haan — examiner training aur ek adjudication standard deployment ki formal requirement hai.

B. Legal / Procedural

Yahan short aur imandaar jawab hi sabse strong hai.

29. Ye court me admissible hai?

System ka output evidence nahi, decision-support hai — admissibility examiner ke conclusion aur uske documentation par tikti hai.

30. Court me challenge hua to aap kaise defend karoge?

Har number apni source artifact file ka naam carry karta hai, aur limitations report me pehle se likhi hain — defend karne ko kuch chhupaya hi nahi gaya.

31. Opposing expert kya attack karega? ■

Sabse pehle photo-anthropometry — isiliye humne wo claim kiya hi nahi; document ka sabse kamzor claim poore document ki ceiling ban jata hai.

32. ENFSI / FISWG guidelines follow karte ho?

Haan — ENFSI 2015 evaluative reporting ke verbal bands aur FISWG ka morphological comparison approach.

33. Likelihood ratio report karte ho?

Engine me calibrated LR aur ENFSI bands built aur tested hain; product report me unki wiring abhi outstanding hai aur ye likha hua hai.

34. Data protection / privacy compliance kya hai?

Encryption at rest, retention window, real erasure, tamper-evident audit trail aur role-based access.

35. DPIA hui hai? ■

Abhi nahi — ye deployment se pehle ki formal requirement hai aur checklist par hai.

36. Data kitne din rakhte ho?

Probe images default 90 din (configurable); enrolment images subject record ke jeevan tak.

37. Kisi ka data delete karna ho to?

Subject record delete karo — templates, enrolment images aur index entries sab hat jate hain.

38. Delete ka matlab sach me delete hai ya hidden?

Sach me delete — koi soft-delete flag nahi; sirf ye record bachta hai ki ek search hui thi.

39. Biometric consent ka kya framework hai?

Consent framework deploying organisation ka hai — system consent nahi, lawful basis record karta hai.

40. Jurisdiction-specific legal review hua hai?

Nahi — wo deployment checklist par explicitly listed hai.

C. Database Engineer

Ye banda schema aur isolation par jayega. Q51 aur Q56 sabse likely.

41. Database kaunsa hai?

SQLite development ke liye; PostgreSQL production ke liye configured aur tested.

42. Kahan host hai?

Abhi local machine par — `runtime/imatch.db`, repo root se anchored.

43. Schema kya hai — kitni tables?

11 — `tenants`, `users`, `api_keys`, `cases`, `subjects`, `templates`, `search_runs`, `candidates`, `enhancement_runs`, `report_narratives`, `audit_records`.

44. Face data kis table me, kis format me?

`templates` table me, `nonce` + `ciphertext` columns — AES-256-GCM se encrypted 512-dimension vector.

45. Photos database me store hoti hain ya disk par?

Disk par — database me sirf path aur SHA-256 hash rehta hai.

46. Photos disk par kaise organise hain?

`storage/<tenant>/<probes|enrolments|enhanced>/ab/cd/<sha256>.jpg` — two-level fan-out, kyunki ek flat directory me lakh files filesystem ke liye takleef hai.

47. Encryption at rest hai?

Haan — `templates` AES-256-GCM se, per-tenant HKDF subkey ke saath.

48. Encryption key kahan rakhi hai?

Environment variable me; set na ho to system loudly warn karke unencrypted store karta hai, jo production me kabhi allowed nahi.

49. Key rotate kaise karoge? ■

Design per-tenant subkey derive karta hai; bulk re-encryption utility abhi roadmap par hai — ye outstanding item hai.

50. Multi-tenancy kaise handle ki hai?

Row-level `tenant_id` ke saath-saath structurally alag vector matrices — sirf filter par nahi.

51. Ek tenant ka data dusre me leak ho sakta hai? ■

Nahi — `search()` tenant ID required argument leta hai, to cross-tenant compare karne ka code path hi maujood nahi.

52. Indexes kaunse hain?

`tenant_id`, `case_id`, `operator_id`, `created_at`, `probe_sha256`, `image_sha256`, aur `audit` par unique (`tenant`, `sequence`).

53. Migrations kaise manage hote hain?

Alembic.

54. Backup strategy kya hai? ■

Abhi file-level manual backup — formal backup aur restore policy production checklist par pending hai.

55. Concurrent writes handle hote hain?

SQLite par WAL mode ke saath; audit chain me unique (tenant, sequence) constraint concurrent append ko silent fork ki jagah loud failure banata hai.

56. SQLite production ke liye theek hai? ■

Nahi — concurrent load nahi sambhalega; ye khud hamari documentation me likha hai aur Postgres path ready hai.

57. Postgres par migrate karne me kya lagega?

Sirf `NEXGEN_DATABASE_URL` set karna — scheme rewrite automatic hai, code change zero.

58. Audit table append-only kaise enforce ki?

Application level par hash chain + sequence uniqueness; DB-level grant restriction deployment ki zimmedari hai.

D. Software / Backend Engineer

Q74, Q78 aur Q75 par sabse zyada dabav aayega.

59. Tech stack kya hai?

FastAPI + SQLAlchemy + Alembic backend; React 19 + Vite 7 + Tailwind frontend; InsightFace + ONNX Runtime GPU + PyTorch engine.

60. Architecture ka overview do

imatch_api (product) aur *nexgen_engine* (framework-agnostic recognition library) alag hain, to engine independently test ho sakta hai.

61. API kaunse endpoints expose karta hai?

auth, account, admin, cases, subjects, search, reports, audit, enhancement, health.

62. Authentication kaise kaam karti hai?

JWT bearer token ya X-API-Key; principal hamesha verified credential se resolve hota hai, request body se kabhi nahi.

63. Password hashing kya use kiya?

Argon2id, OWASP interactive parameters, cost badalne par transparent rehash.

64. Token expiry aur refresh flow?

Short-lived access token + alag refresh token (7 din, remember-me 30) — refresh ko access ki jagah bhejo to reject hota hai.

65. Session revoke kaise hota hai?

session_epoch badhao — us user ke saare live sessions turant invalid.

66. Rate limiting hai?

Haan — 120 requests/minute general, 30/minute search par.

67. File upload validation kya hai?

15 MB ceiling (base64 decode se pehle aur baad me dono), aur format magic bytes se identify hota hai, declared content type se nahi.

68. Bade ya corrupt file par kya hota hai?

Typed 4xx usable message ke saath — 500 nahi.

69. Error handling kaisi hai — 500 aate hain?

Unhandled exception par generic message + request ID; stack trace kabhi expose nahi hota kyunki wo schema aur paths leak karta hai.

70. Logging aur observability kya hai?

Per-stage *StageTimings*, correlation X-Request-ID, structured logs, p50/p95/p99.

71. Latency kya hai per request?

Encode 14.7 ms (p50), full 1:1 verify 31 ms — GPU par.

72. Throughput kitna hai?

Single-threaded ~71/s; 4 workers par 132/s; request batching se 551/s.

73. Concurrency kitni handle hoti hai?

4 workers par saturate — 8 workers kuch nahi dete aur p99 ko 2.3× kharab kar dete hain.

74. Scale karne par kya toot sakta hai? ■

Gallery ~100k se upar search cost probe encode ke barabar ho jata hai — tab approximate index chahiye, aur uska recall loss measure karke accept karna padega.

75. Test coverage kitna hai?

297 tests pass — 144 engine + API + 23 adversarial — plus ek regression gate jo model/threshold/fusion change par poori suite force karta hai.

76. CI/CD setup hai?

GitHub Actions CI hai; CD nahi, kyunki abhi koi hosted environment hi nahi hai.

77. Deployment kaise hoga?

Docker (port 8443) + Render blueprint ready hai; Postgres, TLS ingress aur CORS restriction deployment checklist par hain.

78. Abhi live hai? ■

Nahi — locally verified hai, hosted backend abhi nahi, aur ye ek documented blocker hai.

79. Docker / container support hai?

Haan — Dockerfile aur docker-compose dono, single-origin self-host stack samet.

80. Frontend kaise banaya hai?

React 19 + Vite + Tailwind; alag workspace pages — Search, Verify, Enrol, Batch, Cases, Audit, Enhance.

81. GPU chahiye ya CPU par bhi chalega?

Dono chalte hain — CPU par sab kaam karta hai, bas benchmarks kaafi dheere.

E. Security Engineer

Q84, Q92 aur Q95 par imaandari hi sabse achha jawab hai.

82. Threat model kya socha hai?

Insider misuse, cross-tenant leak, template theft, audit tampering, hostile upload aur SSRF.

83. Biometric template leak ho jaye to?

Biometric irrevocable hai — password ki tarah reissue nahi hota; isliye template ko photo ke barabar treat kiya jata hai aur encrypted rakha jata hai.

84. Template se face wapas ban sakta hai? ■

Haan, ek pehchane-jaane layak approximation ban sakta hai — isliye API sirf metadata deta hai, raw embedding kabhi nahi.

85. Template API se bahar jata hai?

Kabhi nahi.

86. SQL injection / XSS / CSRF protection?

Parameterised ORM queries, CSRF protection, security headers — koi raw SQL nahi.

87. SSRF ka kya handling hai?

Server-side URL import jaan-boojh kar 501 return karta hai, kyunki wo internal addresses aur cloud metadata endpoints tak pahunchne ka primitive hai.

88. Security headers kaunse lagaye hain?

Cache-Control: no-store, X-Content-Type-Options, X-Frame-Options: DENY, Referrer-Policy: no-referrer, X-Request-ID, aur production me HSTS.

89. Secrets management kaise hai?

Environment variables; .env kabhi commit nahi hota; startup par validation hoti hai.

90. JWT secret set na ho to kya hota hai? ■

Ephemeral per-process secret ban jata hai aur restart par saare tokens mar jate hain — ye ek real bug tha jo pakda aur fix hua.

91. Template key set na ho to? ■

Templates unencrypted store hote hain — system loudly log karta hai, aur production me ye kabhi allowed nahi.

92. Penetration test hua hai? ■

External pen-test nahi — internal adversarial testing hui hai. Ye ek declared gap hai.

93. Insider threat ke against kya control hai?

Hash-chained audit, har user ko apna trail padhne ka access, role separation, aur enrolment par supervisor requirement.

94. Adversarial input testing ki hai?

Haan — 23 tests: corrupt bytes, zero-byte files, extreme aspect ratios, non-face images.

95. Audit log tamper-proof kaise hai, aur kya prove nahi karta? ■

Chain prove karti hai ki kuch **badla nahi**; wo **completeness prove nahi karti** — uske liye mirrored JSONL ko write-once storage par bhejna padega.

F. AI / ML Engineer

Sabse deep section. Q97, Q104, Q107, Q116, Q122 par tayyari sabse zaroori.

96. Kaunsa model use kar rahe ho?

ArcFace w600k_r50 (ResNet-50, WebFace600K) deployed; ViT-B + KP-RPE + AdaFace evaluate hokar adoption ke liye approved.

97. Model khud train kiya ya pretrained? ■

Pretrained — clean-image parity isliye hai kyunki wahi state-of-the-art hai; hamara contribution measurement discipline aur capacity framework hai.

98. Training data kya thi?

Pretrained weights WebFace600K par; hamara LoRA fine-tune TinyFace + QMUL training splits par — 63,813 images / 7,889 identities.

99. Embedding dimension kya hai?

512-d, L2-normalized.

100. Similarity metric kya hai?

Cosine similarity — normalized vectors par dot product ke barabar.

101. Threshold kaise decide kiya?

FMR = 0.1% ke rule par, AgeDB-30 ke 12,000 pairs se — accuracy maximise karke nahi.

102. Threshold kis dataset par calibrate hua?

AgeDB-30 par, within-subgroup matched impostors ke saath — same gender, same age band.

103. FAR / FRR / TAR kya hai aapka?

Deployed point par FAR 0.10%, FRR 6.32% overall — aur under-25 me FRR 14.75%.

104. TAR@FAR quote karte ho ya sirf accuracy? ■

TAR@FAR — akela TAR meaningless hai, kyunki threshold zero karke use 100% banaya ja sakta hai.

105. Benchmark kaunse chalaye?

LFW, CFP-FF, CFP-FP, AgeDB-30, CALFW, CPLFW, TinyFace, QMUL-SurvFace, SCface, IJB-B, IJB-C.

106. IJB protocol official use kiya?

Haan — 5-point norm_crop, media-aware template pooling, official pair lists; IJB-B 95.44% aur IJB-C 96.91% at FAR=1e-4.

107. Train / test data overlap audit kiya? ■

Haan — teeno training archives me overlap mila (692 of 10,572 CASIA identities excluded), aur fine-tune data zero overlap par audit hua.

108. 10-fold cross-validation kaise kiya?

Threshold 9 folds par fit hota hai aur held-out 10th fold par apply hota hai — test data par fit kabhi nahi.

109. Degraded imagery par performance kyun girti hai?

25–32 pixel face me identity information hi kam hai — humne naapa: ~2.9 bits available vs ~11.5 bits required.

110. Fine-tuning try ki? Result kya raha?

Do baar — synthetic degraded data par fail (har benchmark kharab), real low-res data par LoRA se pass (QMUL bits 2.92 → 4.79).

111. LoRA kyun, full fine-tune kyun nahi? ■

6 GB VRAM me ViT-B ka optimiser state usable batch size par fit nahi hota; LoRA 1.2% parameters train karta hai aur catastrophic forgetting bhi kam karta hai.

112. Backbone swap kiya to kya mila?

TinyFace TAR@FAR0.1% 20.59% → 60.77% (3×), rank-1 32.93% → 68.20%; clean packs par 0.03–0.25 points ka chhota cost, jo humne round off nahi kiya.

113. Ensemble kyun nahi use kiya? ■

Measure kiya — 3-model ensemble 5 me se 4 datasets par best single model se hara, 3× compute par.

114. FAISS / ANN search kyun nahi lagaya?

Measured — 10k tak jeetne ko kuch nahi (search encode ka 8% hai), aur mojudaa branch exact index hai, approximate nahi.

115. Multi-frame fusion try ki? ■

Haan, aur wo counterproductive nikli — open-set AUC 1 frame par 0.5399 se 16 frames par 0.4785, yaani chance se neeche.

116. Demographic bias measure kiya? ■

Haan — women 1.7× FNMR, under-25 3.8×; threshold badalne se errors shift hue, gap gaya nahi.

117. Model drift monitor karte ho?

Regression gate model/threshold change par chalti hai; live production drift monitoring abhi pending hai.

118. Explainability kya hai?

Report me comparison plates, per-stage timings, processing plan trace aur skip reasons; model saliency roadmap par hai.

119. Capacity / identity-bits framework kya hai?

Ek observation kitni identity bits deta hai vs gallery kitni maangti hai — isne backbone swap ka result **pehle se predict** kiya tha aur swap ne confirm kiya.

120. Open-set problem ko kaise handle karte ho? ■

Ye hamari sabse badi weakness hai — TPIR 0.00% at 1% FPIR; LoRA ke baad 1.38%. Isiliye human adjudication mandatory hai.

121. Enhancement se accuracy badhi? ■

Nahi — SCface par har enhancement arm reject hua, interpolation control ko bhi nahi hara paaya. Enhancement sirf visual examination ke liye hai.

122. Apne code me koi measurement bug mila? ■

Haan — 12 defects, jinme se 4 ne published numbers produce kiye the; sab documented hain symptom, cause aur fix ke saath.

123. Koi published number withdraw kiya? ■

Haan — TinyFace rank-1 37.43% → 32.93%, QMUL rank-1 0.53% → 2.68%, aur do capacity tables poori withdraw.

124. Hardware kya use kiya? ■

Ek RTX A3000 Laptop, 6 GB — poore project ka binding constraint yahi raha, aur usi ne LoRA vs full fine-tune jaisi har choice decide ki.

125. NIST FRTE submit kiya ya karoge?

Abhi nahi — Phase 8 (NIST readiness) roadmap par hai, FRTE 1:1 primary submission ke saath.

G. Product / Management

Yahan short, confident aur honest rehna hai.

126. Ye kis problem ko solve karta hai?

Investigator ke paas kharab CCTV image hai aur use pata karna hai ki dekhne layak kaun hai.

127. Competitor se kaise alag hai? ■

Accuracy se nahi — honesty se: claims condition-wise separate hain, structurally constrained hain, aur galat nikalne par public me correct hote hain.

128. Kitna time laga banane me?

Lagbhag paanch hafte — 1 July se 5 August 2026, 98+ commits.

129. Team size kitni thi?

Chhoti — ek lead engineer.

130. Deployment cost kya hoga?

GPU host + Postgres + object storage; free tier par nahi chalega kyunki model load hi 350 MB+ hai.

131. Kitne users / gallery size support karega?

Gallery 10k tak comfortably; 100k par search probe encode jitna mehnga ho jata hai.

132. Roadmap kya hai?

Phase 8 NIST readiness → 9 measured wins ship → 10 degraded imagery → 11 submit → 12 product completeness → 13 publish.

133. Kya kaam abhi baaki hai? ■

Backend hosting, evidence layer ki report wiring, independent validation, aur video ingestion.

134. Sabse bada risk kya hai? ■

Independent validation ka na hona — koi bhi internal kaam use band nahi kar sakta.

135. Independent validation kab karvaoge? ■

Phase 11 me NIST FRTE submission ke through — result jo bhi ho, bina spin ke publish hoga.

H. Killer questions — inhe sabse pehle ratto

Ye saat sawal poori meeting ka rukh badal sakte hain.

136. Aapne khud model train kiya ya open-source utha liya? ■

Pretrained ArcFace weights — aur main ye chhupa nahi raha: hamara contribution recognition model nahi, measurement aur validation discipline hai.

137. Ye numbers kisne verify kiye? ■

Koi nahi — sab kuch hamari hi team aur tooling se, public academic datasets par. Ye hamara sabse bada gap hai aur documentation me likha hua hai.

138. Live demo dikhao abhi ■

Local par abhi dikha sakta hoon — hosted deployment abhi nahi hai, aur wo ek documented blocker hai.

139. Paanch saal purani photo se match kar sakte ho? ■

AgeDB-30 par 98.15%, par 30-saal ke span par under-25 group me false-reject 14.75% tak chala jata hai.

140. Aapka system kabhi galat ho sakta hai — haan ya na? ■

Haan — aur exact rate bata sakta hoon: lagbhag 1,000 comparisons me 1 false match.

141. Aap guarantee de sakte ho ki ye sahi aadmi hai? ■

Nahi. Ye ek lead hai, identification nahi — guarantee dena is system ka kaam hi nahi hai.

142. Ye AI se banaya hai kya? ■

System ka design, measurement protocol, decision rules aur validation mera hai — har number ke saath uska artifact file naam likha hai jise koi bhi khud verify kar sakta hai.

Annex — Architecture layer cheat sheet

Agar koi poochhe “kis layer me kya use kiya” — ye table dikha dena.

Layer	Kya use hua	Key detail
1. Face detection	SCRFD (det_10g, buffalo_l pack), ONNX Runtime GPU, det_size 640×640	Chhote face ke liye padded retry; GPU binding assert hoti hai, silent CPU fallback impossible
2. Face alignment	5-point similarity transform (Umeyama) → 112×112 norm_crop; plus DFA aligner ViT path ke liye	Surveillance crops par DFA mandatory — bina iske TinyFace ~51 points gir jata hai. Landmarks se pose bhi nikalta hai
3. Image quality	ImageQualityFilter — 8 components, blocking vs advisory alag; plus blur σ , noise σ , JPEG quality estimation	Sharpness = Laplacian variance; teeno degradation estimates confidence-gated; effective_cutoff se true resolution
4. Image enhancement	Track A: 6 classical operators (deblock, bounded deblur, CLAHE, denoise, Lanczos). Track B: Real-ESRGAN, NAFNet, CodeFormer, GFPGAN	Track B default OFF; weights checksum-pinned; 6 GB card ke liye sequential load→apply→release with per-stage VRAM measurement
5. Feature extraction	ArcFace w600k_r50 (ResNet-50, WebFace600K) → 512-d L2-normalized; alternatives ViT-B KP-RPE AdaFace aur ViT+LoRA r8	Flip-TTA sirf horizontal mirror, kyunki ArcFace flip augmentation ke saath train hua hai
6. Face database	SQLite / PostgreSQL via SQLModel + Alembic; templates AES-256-GCM per-tenant HKDF; images content-addressed on disk	In-memory gallery index DB se rebuild hoti hai, per tenant, first use par. Photos DB me nahi jati
7. Face matching	Cosine similarity = dot product on L2-normalized vectors; exhaustive numpy matmul (exact)	Tenant-sharded — tenant ID required argument. FAISS IndexFlatIP branch dormant. margin aur z-norm statistics available
8. Decision engine	ThresholdConfig single source — match 0.2871, review 0.2153, verify 0.2871	Teen band: match / review / no-match. Quality, liveness aur deepfake advisory score me fuse hote hain, decision override nahi karte. Final decision hamesha human

Ek line me poora flow

Image → SCRFD detect → 5-pt align 112×112 → quality gate → [optional enhance] → ArcFace 512-d embedding → cosine vs gallery → threshold band → HUMAN adjudication

Paanch numbers jo zubaani yaad hone chahiye

Threshold 0.2871 / 0.2153 · **FAR** 0.10%, **FRR** 6.32% · **LFW** 99.78%, **TinyFace** 20.59% → 60.77%, **QMUL** 2.31% · **IJB-B/C** 95.44% / 96.91% at FAR=1e-4 · **Speed** encode 14.7 ms, verify 31 ms